



VLSI Testing

積體電路測試

Design For Testability (DFT)

Part I: Internal Scan Chains

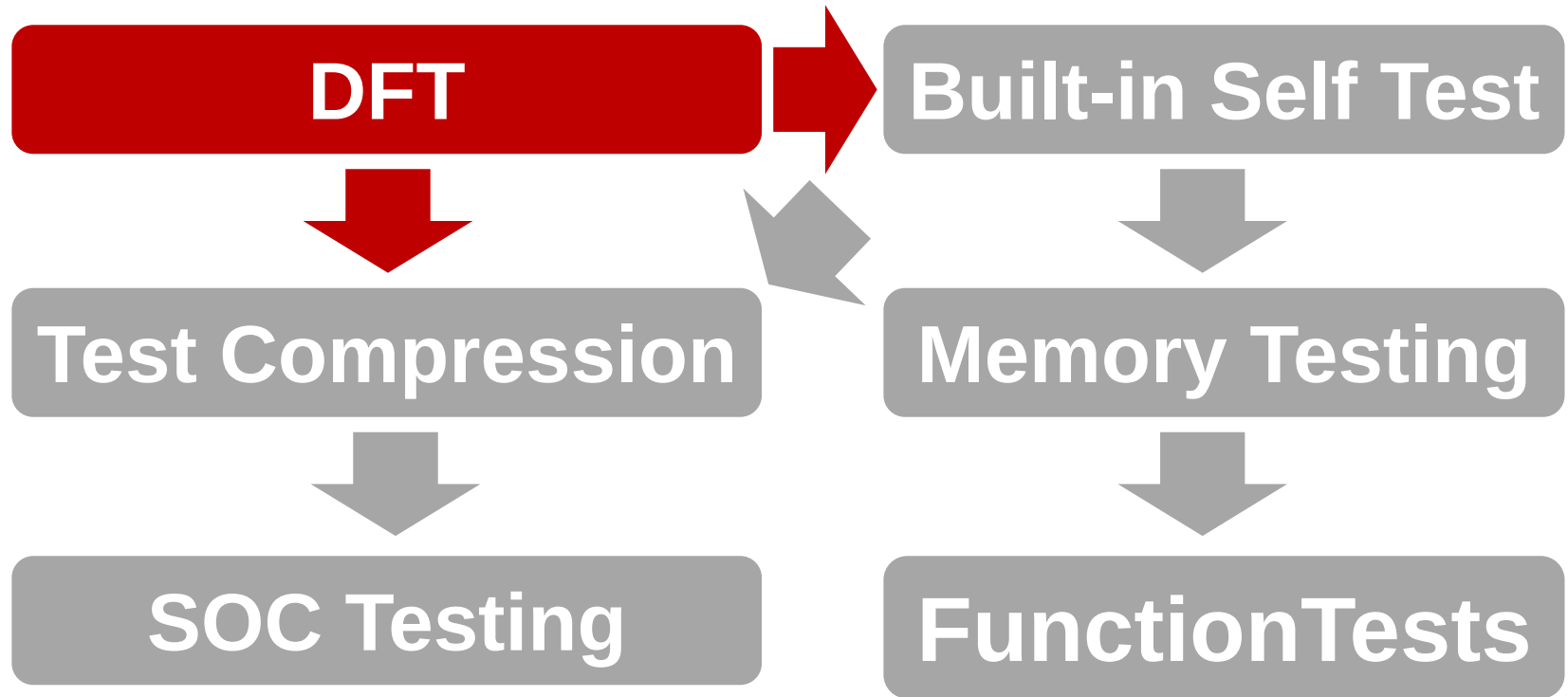
Professor James Chien-Mo Li 李建模

Lab. of Dependable Systems

Graduate Institute of Electronics Engineering

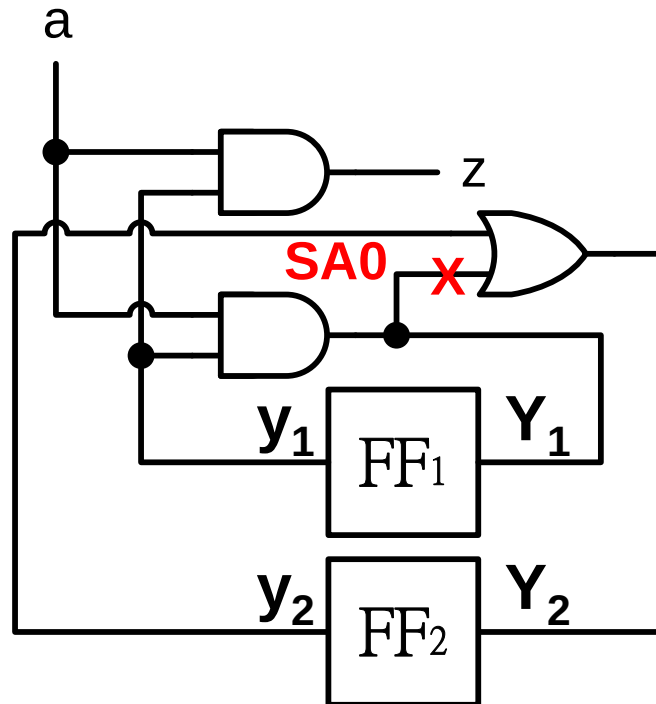
National Taiwan University

Course Roadmap (Design Topics)



Motivating Problem

- Sequential ATPG fails to generate test pattern. Your manager asked you to **add circuits** so that this fault can be detected.
- Can we change FF to improve controllability/observability?



Why Am I Learning This?

- **Design for Testability**

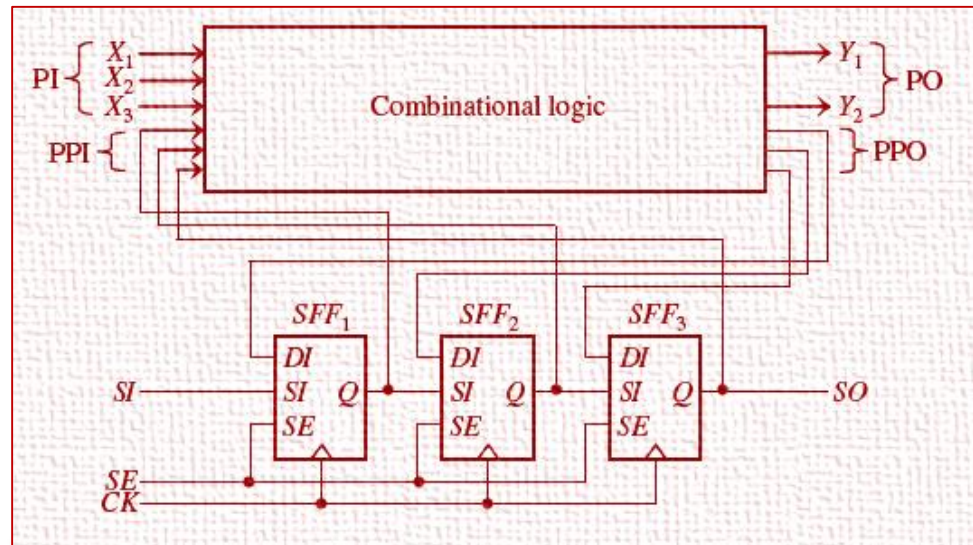
- ◆ Make ATPG easier
- ◆ Improve fault coverage (test quality)
- ◆ Reduce test length (test cost)

*“Good design is obvious.
Great design is transparent.”*

(Joe Sparanolf)

DFT Outline

- Part 1
 - ◆ Introduction
 - ◆ Internal Scan
 - * FF-based: MUX-scan, clock scan
 - * Latch-based: LSSD
- Part 2
 - ◆ External Scan
 - * JTAG (1149.1)



Design for Testability (DFT)

• What is DFT?

- ♦ Insert circuitry that has little to do with function, but
- ♦ Improves **testability** : controllability and observability

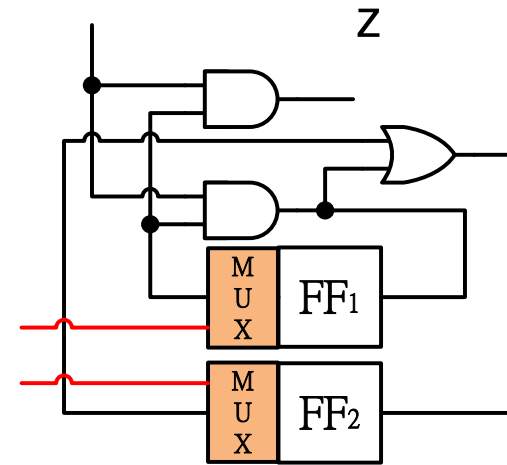
• Why DFT?

- ♦ 1. Reduce **ATPG effort**
 - * Shorten ATPG run time
- ♦ 2. Improve **test quality**
 - * High fault coverage
- ♦ 3. Reduce **test cost**
 - * Shorten test length and test time
- ♦ 4. Reduce **time to market**
 - * Easy debug and diagnosis



Penalty of DFT

- 1. **Design effort overhead**
- 2. **Performance degradation**
- 3. **Hardware overhead**
 - ♦ extra area, extra pins
- 4. **Yield loss**
 - ♦ Larger area means lower yield
- 5. **Power overhead**
 - ♦ “Power tax” of testing
- In 1970's, expensive silicon and simple design, DFT not popular
 - ♦ Now, cheaper silicon and complex design, DFT is necessary



DFT is Necessary for Modern Designs

Quiz

Q: Which of the following is NOT true about DFT?

- A. DFT reduces designers' effort**
- B. DFT cost extra hardware and power**
- C. DFT reduces test cost so it is needed for complex design**

ANS:

DFT - Part 1

- **Part 1**

♦ Introduction

- * Ad-hoc DFT 
- * **Test point insertion**

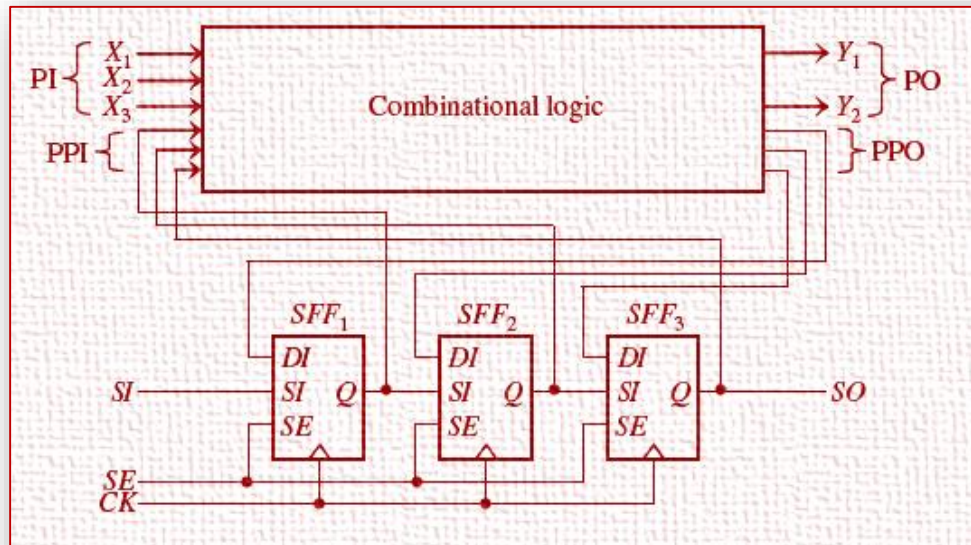
◆ Internal Scan

- * FF-based
- * Latch-based

● Part 2

◆ External Scan

- ## * JTAG (1149.1)



Ad Hoc DFT

- What is “Ad hoc” ?

- ◆ for the particular end or case at hand without consideration of wider application [Webster dictionary]

- What is *Ad Hoc DFT* ?

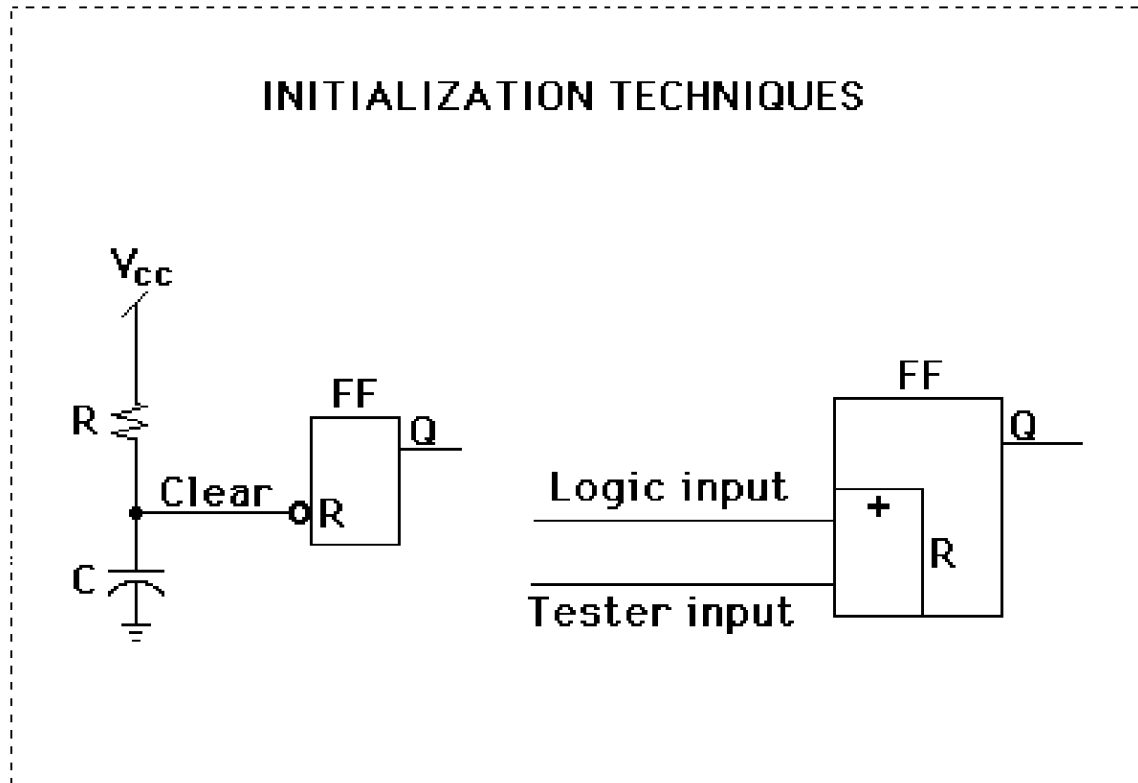
- ◆ DFT technique for a particular design
 - * Not generally applicable to all designs
- ◆ Relies on **good design practice** learned from **experience**

- Ad-hoc DFT examples:

- ◆ Rule1: design circuits easily initializable
- ◆ Rule2: disable internal clocks during test
- ◆ Rule3: partition large circuits into small blocks
- ◆ Rule4: insert test points into circuits of low testability
- ◆

Ad Hoc DFT Example (1)

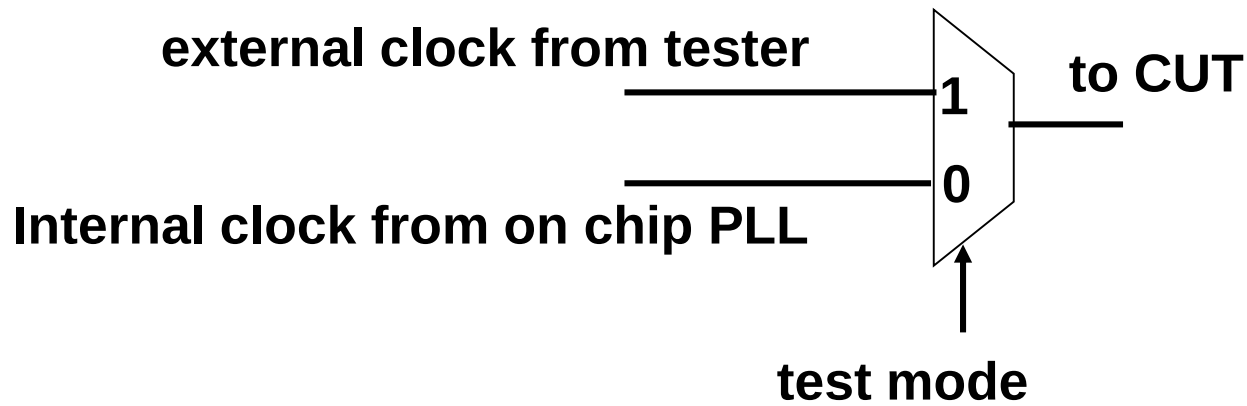
- Rule1: design circuits easily initializable
 - ◆ Self-initialization, or
 - ◆ Initializable from tester



Courtesy Prof. McClueksy, Stanford University

Ad Hoc DFT Example (2)

- Rule2: disable internal clocks during test
 - ◆ Disable on-chip **PLL** in test mode



DFT - Part 1

- Part 1

- ◆ Introduction

- * Ad-hoc DFT

- * Test point insertion

- ◆ Internal Scan

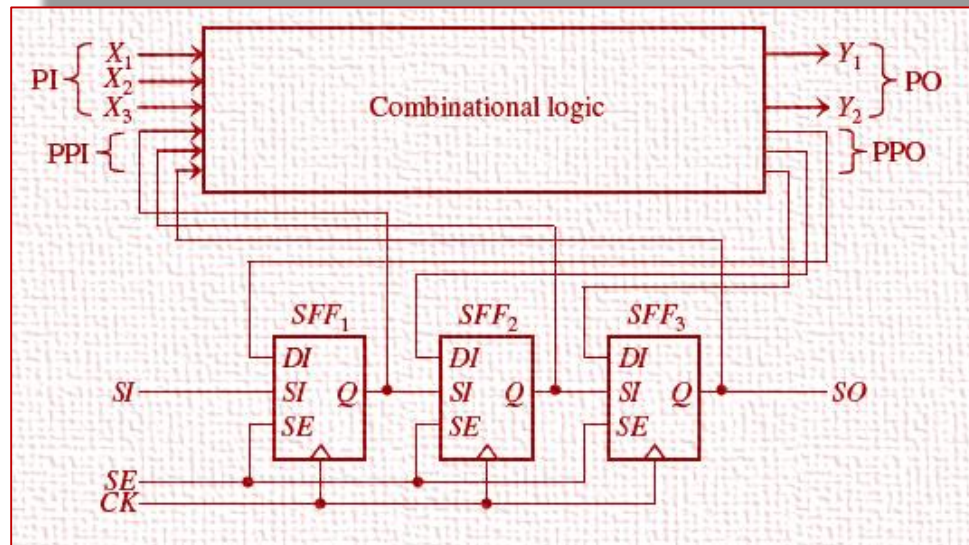
- * FF-based

- * Latch-based

- Part 2

- ◆ External Scan

- * JTAG (1149.1)



Test Point Insertion

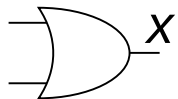
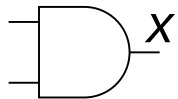
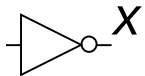
- Objective:
 - ◆ Insert test points to enhance
 - * Controllability
 - * Observability
- Two types of test points
 - ◆ **Control points**: enhance controllability.
 - ◆ **Observation points**: enhance observability

Control Points

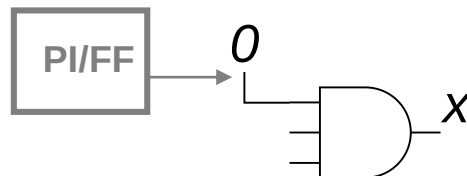
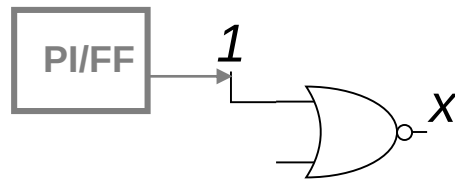
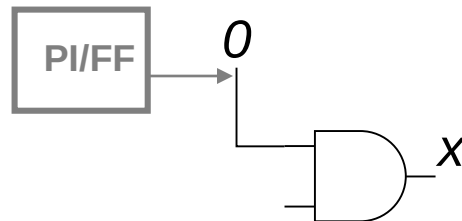
- Control signals come from primary input (PI) or scan FF
 - Details see next video

original

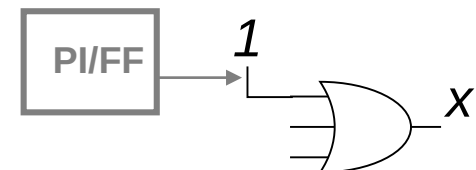
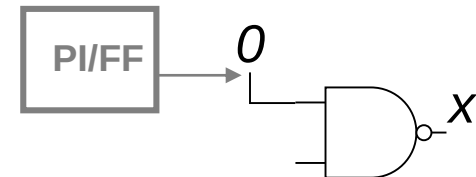
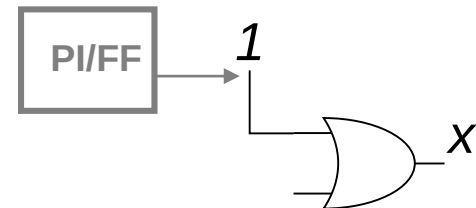
X



control x to 0

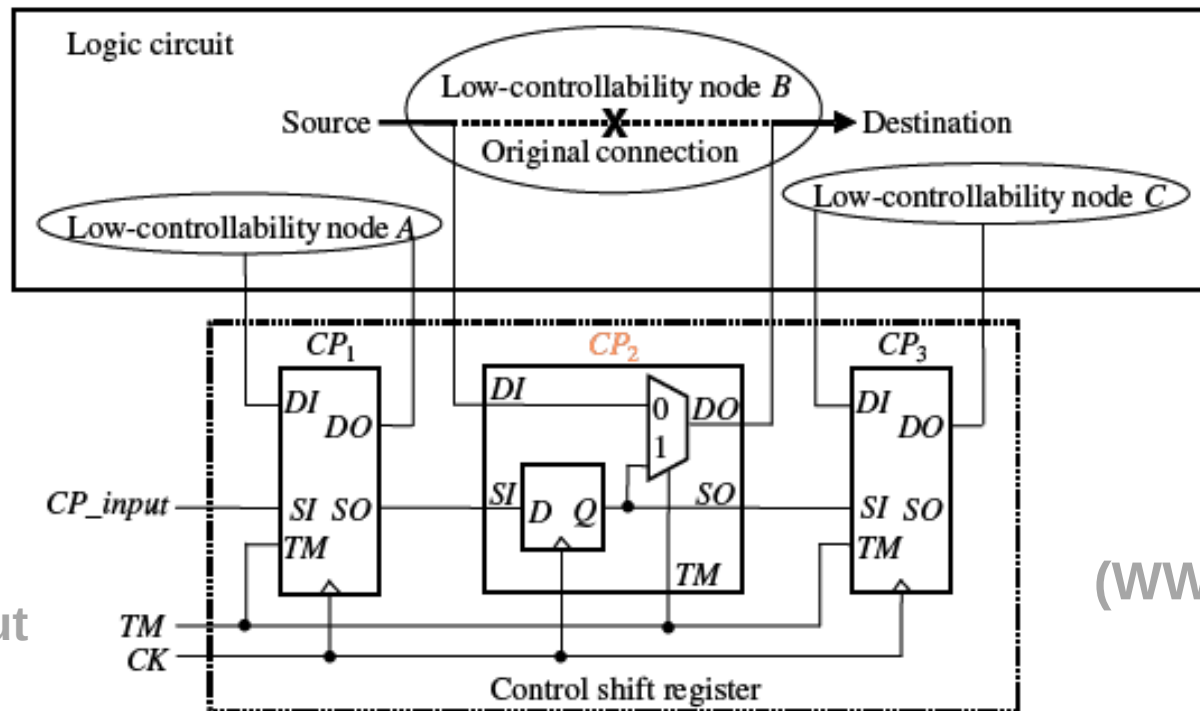


control x to 1



Control Point (2)

- When **TM=0**, normal operation
- When **TM=1**, nodes ABC are controlled by FF 1, 2, 3, respectively



(WWW Fig. 2.5)

DI=data input
DO=data output
TM=test mode
SI=scan input
SO=scan output

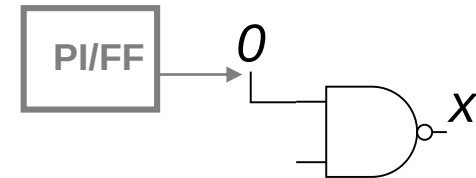
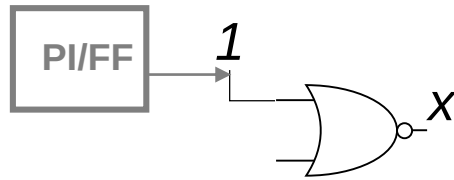
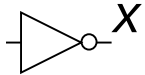
Control point insertion

Quiz

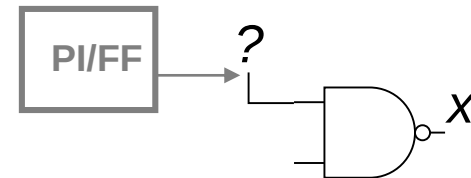
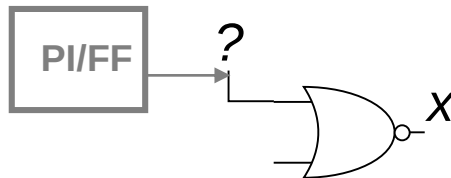
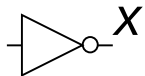
Q: What are control values in NORMAL mode?

ANS:

Test mode

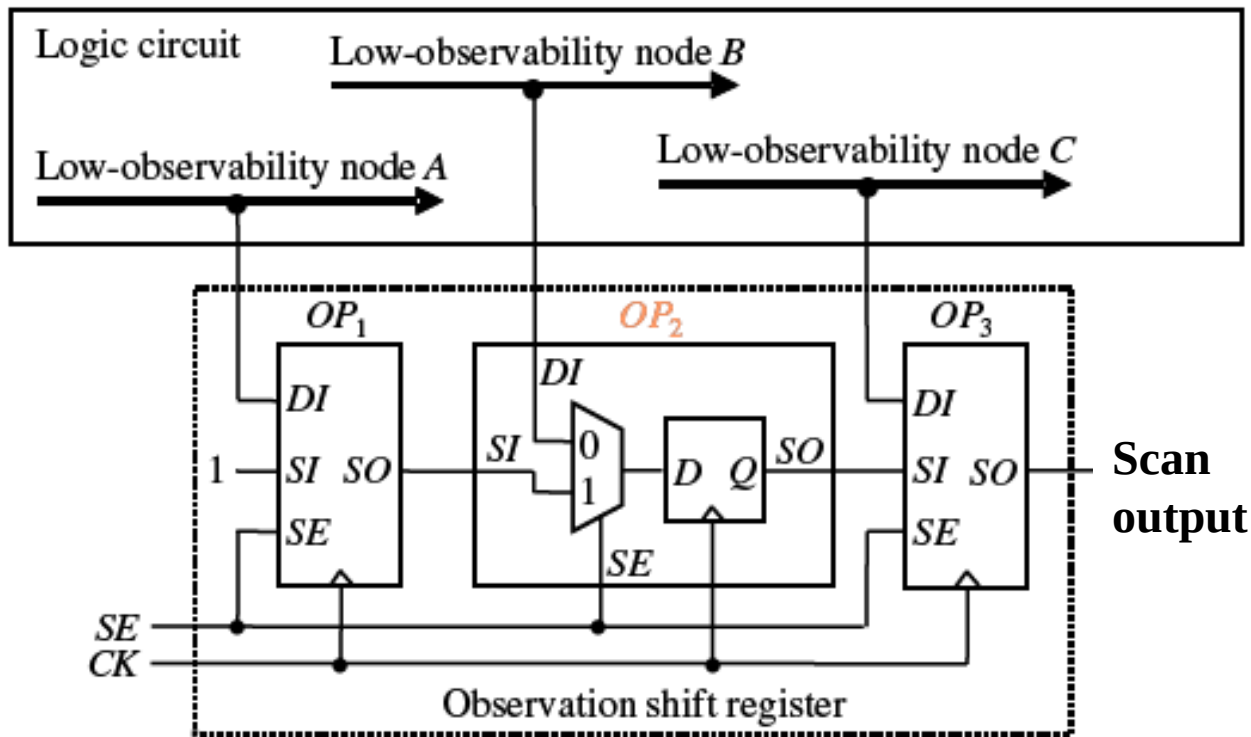


Normal mode



Observation Points

- **SE=0**, ABC are captured into FF by one CK
- **SE=1**, after three CK, values are observed at scan output (SO)

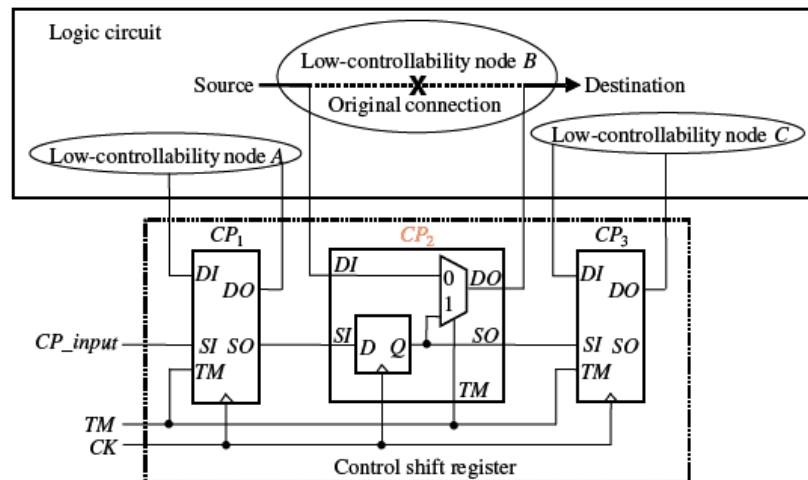


DI=date input
DO=data output
SE=scan enable
SI=scan input
SO=scan output
(WWW Fig. 2.6)

Observation point insertion

Issues with Test Points

- 1. Performance degradation
 - ♦ design slower with test points
- 2. Area overhead
 - ♦ About 1 test point every 1K gates
- 3. Pin overhead
- 4. Where to insert test points?
 - ♦ Need testability measure
- 5. Built-in Self Test (BIST) needs test point more than ATPG
 - ♦ Improve fault coverage , see BIST chapter



Summary

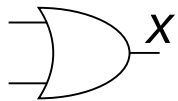
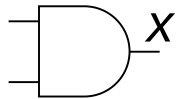
- Introduction to DFT
 - ◆ DFT helps to improve **controllability/observability**
 - ◆ **Ad-hoc DFT** based on experience
 - * Advantage
 - Suitable for your particular design under test
 - * Disadvantage
 - Not systematic → **lacks EDA tool**
 - As opposed to **structured DFT**
 - ◆ Test point insertion
 - * control point/ observation points
 - * **EDA tools ready** for use

Structured DFT is Preferred than Ad Hoc DFT

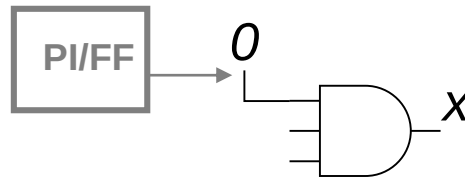
FFT

- Q: What do we do to control AND gate output to 1?

original



control x to 0



?

control x to 1

?

